

Which Retain Set Matters for LLM Unlearning?

A Case Study on Entity Unlearning

Hwan Chang

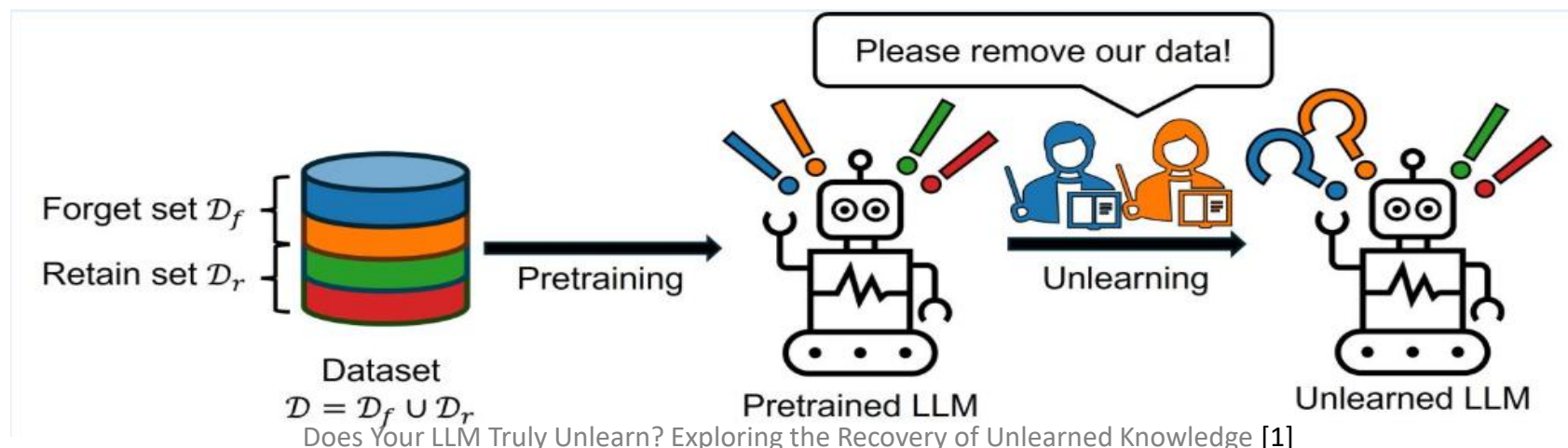
Hwanhee Lee



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Background

- Unlearning aims to: (1) forget information in the forget set and (2) maintain performance on the retain set.
- Prior work focuses on forgetting methods or on verifying whether the forget set has indeed been forgotten.
- However, practical deployment of unlearning is challenging due to retain set performance degradation.



Contribution

- We identify which parts of the retain set are most vulnerable to performance degradation during entity unlearning.
- We demonstrate that syntactic similarity between the forget set and the retain set is a key factor driving degradation in the retain set.
- We show that using syntactically similar data for regularization preserves performance, especially in syntactically similar cases.



Entity Unlearning

Goal: Remove knowledge related to specific entities (e.g., J.K. Rowling, Stephen King) from the model to address privacy concerns.

Data Representation: Each entity is represented as a collection of question-answer (QA) pairs that capture its key information.

J. K. Rowling

Q: When was J. K. Rowling born?
A: July 31, 1965

Stephen King

Q: When was Stephen King born?
A: September 21, 1947



Neighbor Sets

The retain set is too large to analyze thoroughly, so we focus on its subsets—neighbor sets—linked to the forget set

Entity Neighbor Set: QA pairs pertaining to entities that exhibit close associations with the entities in the forget set.

Domain Neighbor Set: QA pairs concerning entities that belong to the identical professional domain as the target entities found in the forget set.

Syntactically Similar Neighbor Set: This set is constructed by grouping questions that exhibit similar syntactic structures to those present in the forget set

Entity	Question	Answer
J. K. Rowling	When was J. K. Rowling born?	July 31, 1965
	Which book concludes the Harry Potter series written by J. K. Rowling?	Harry Potter and the Deathly Hallows
Stephen King	When was Stephen King born?	September 21, 1947
	Which Stephen King novel features a killer clown named Pennywise?	It

(a) Examples of Forget Set.

Neighbor Set Type	Entity example	Example Question	Example Answer
Domain Neighbor Set	Mark Twain (a writer, of the same profession)	On which river did Mark Twain work as a steamboat pilot?	Mississippi River
Entity Neighbor Set	Emma Watson (the lead actress in Rowling's works)	What was Emma Watson's first lead role?	Harry Potter
Syntactically Similar Neighbor Set	When was X born? (similar syntactic structure as in the forget set)	When was Nelson Mandela born?	July 18, 1918

(b) Examples of Types of Neighbor Sets.



Dataset Consturction

- We constructed datasets for two scenarios: a **real-world scenario** and a **fictional scenario (TOFU [2])**.

Real-world scenario

Forget set: Selected 10 prominent figures across various professions.

Entity neighbor set: Selected entities with bi-directional Wikipedia links and co-mentions.

Domain neighbor set: Selected individuals in the same professional domain.

Syntactically similar set: Collected questions sharing similar structures with the forget set.

TOFU

Forget set: Applied a 1% forget ratio to determine the number of fictitious authors.

Entity neighbor set: No entity neighbor set available.

Domain neighbor set: The original dataset only contains domain neighbor sets.

Syntactically similar set: Constructed by clustering questions with similar structures.



Experimental Setup

Models

Real-World Scenario: LLaMA-3-8B-Instruct

TOFU Benchmark: LLaMA-2-7B-Chat

Metrics

Model Utility (MU): Mean of four metrics on retain set

(ROUGE-L Recall, BERT cosine similarity, Probability, Entailment score)

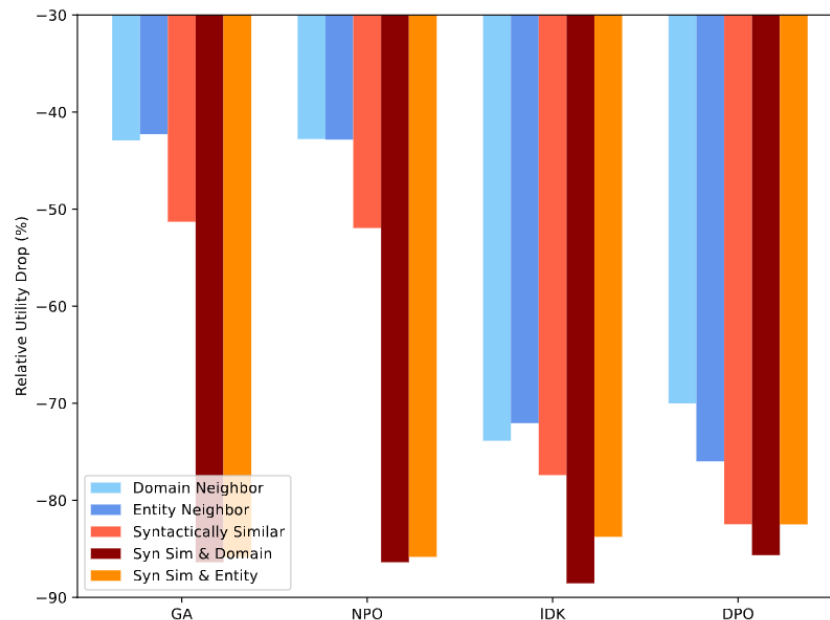
Relative Utility Drop (RUD): Quantifies the percentage decrease in MU caused by unlearning.

$$\text{RUD} = \frac{MU_{\text{after}} - MU_{\text{before}}}{MU_{\text{before}}} \times 100.$$

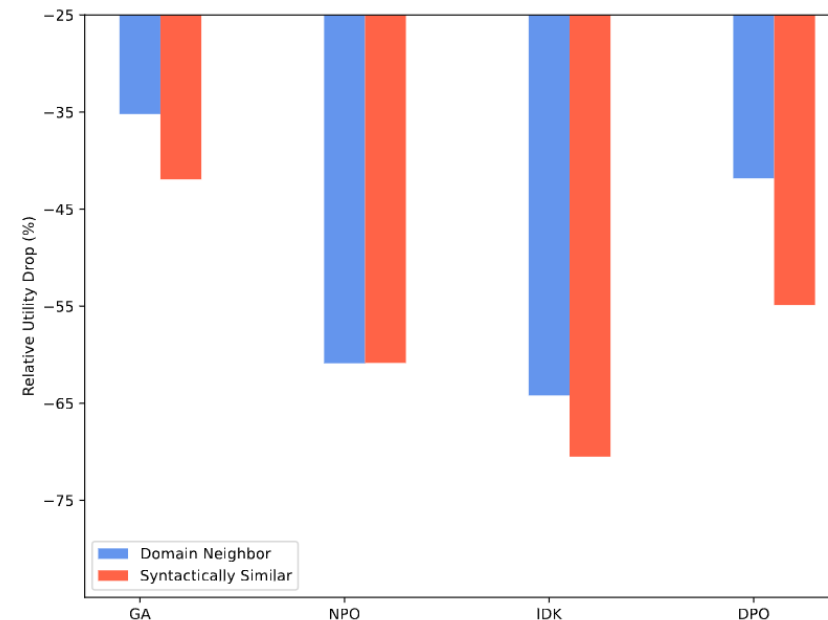


Performance Drops Across Neighbor Sets

- Syntactically similar neighbor set experiences higher forgetting.
- No significant difference among existing neighbor sets.
- Overlapping sets lead to even greater forgetting.



(a) Real-world Scenario



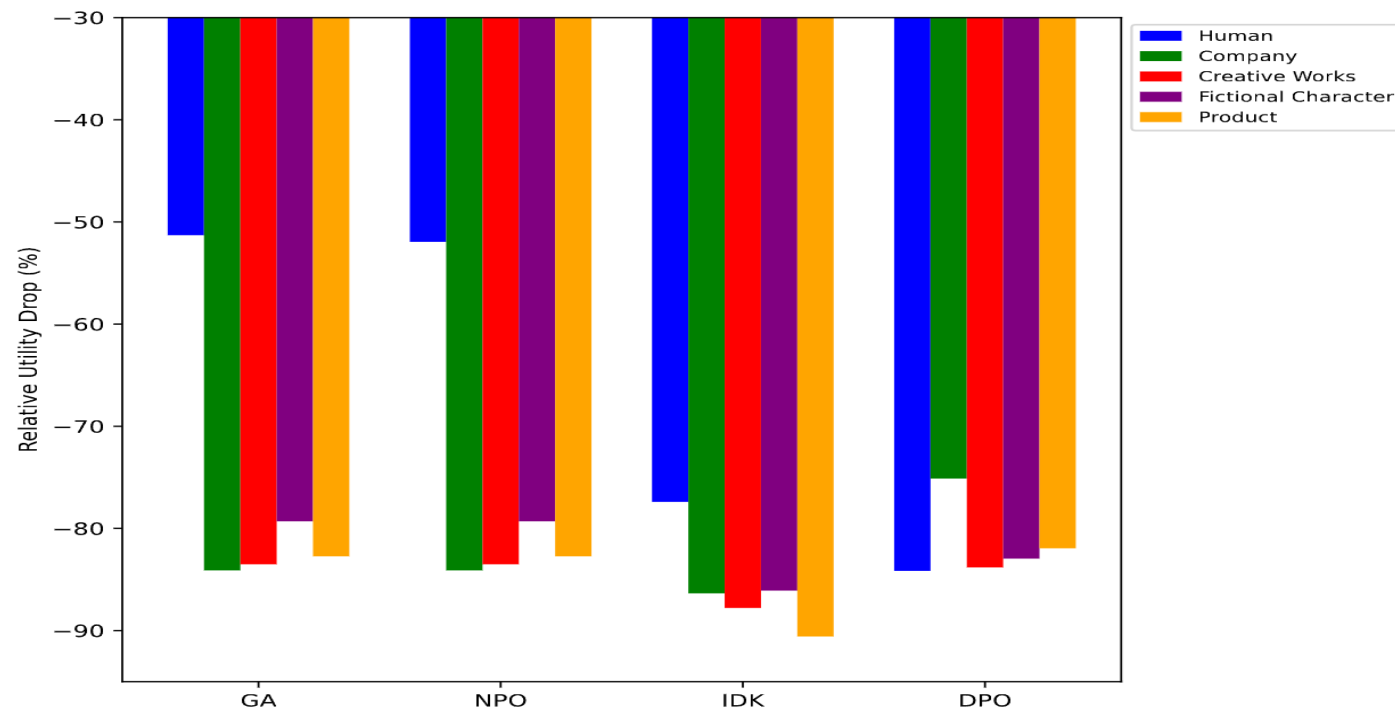
(b) TOFU



Domain Effects on Forgetting in Syntactically Similar Cases

Purpose: To analyze how syntactically similar cases affect unlearning performance across different real-world entity categories.

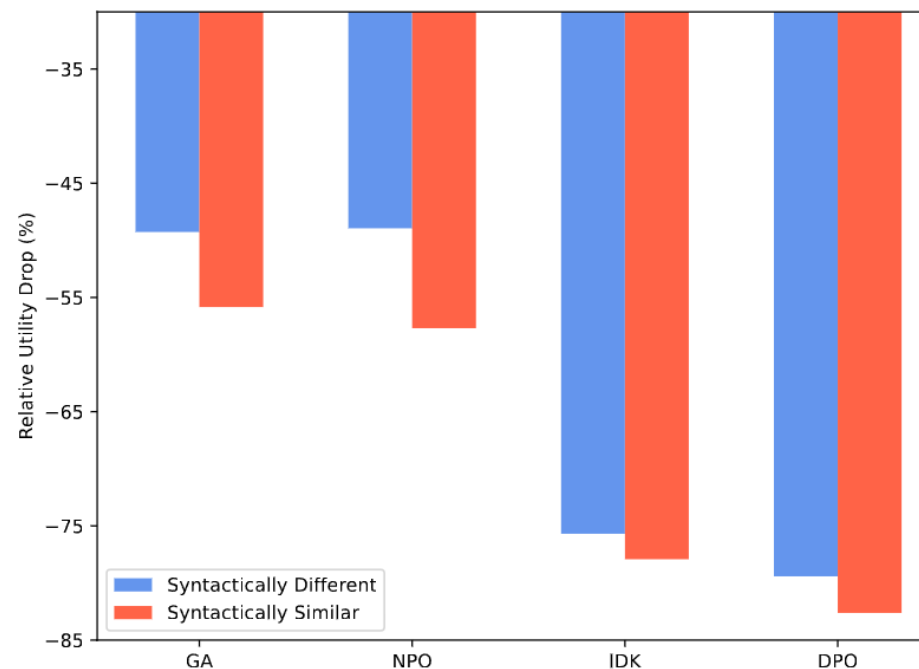
Results: Found that non-human categories exhibit higher forgetting rates, challenging the assumption that entity/domain similarity is the primary driver of performance degradation.



Robustness of Forgetting Patterns in Paraphrased Scenarios

Purpose: To evaluate whether the higher forgetting rates in syntactically similar neighbor sets persist even after paraphrasing.

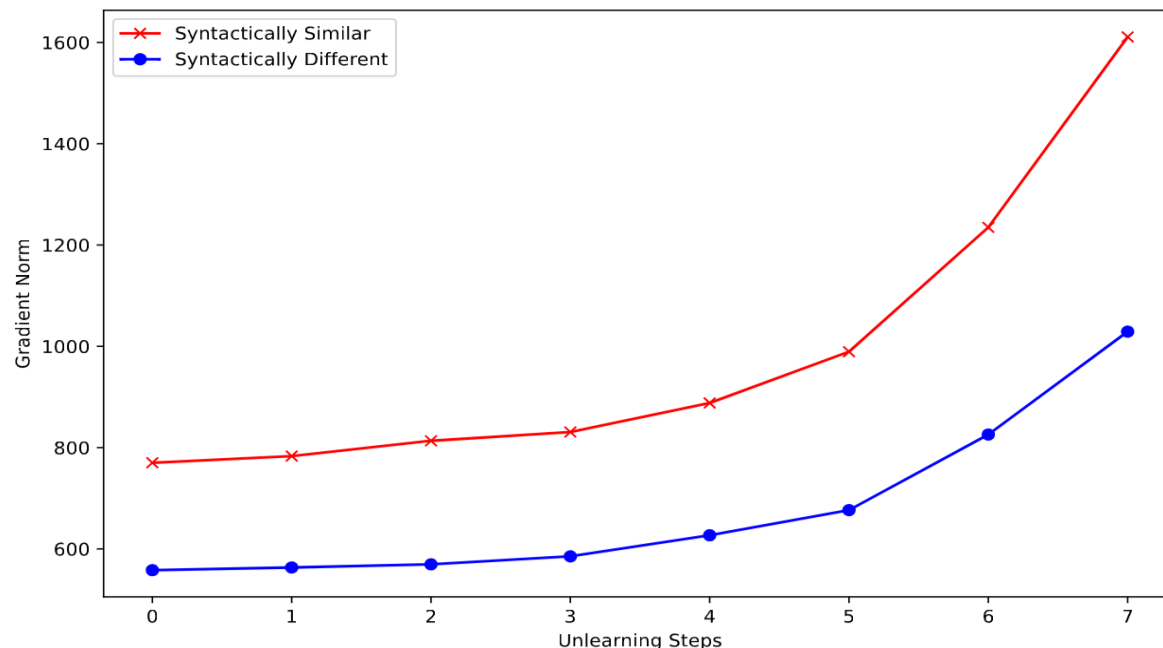
Result: Paraphrased questions still show greater utility drops in syntactically similar neighbors, confirming the robustness of forgetting effects beyond surface-level syntax.



Gradient Analysis

Purpose: To understand how gradient norms evolve during unlearning, particularly whether syntactically similar instances amplify gradient updates compared to syntactically different instances.

Result: Syntactically similar instances show a steeper increase in gradient norms over unlearning steps, widening the gap from dissimilar instances.



Optimal Neighbor Set for Effective Regularization

Regularization losses on the retain set are used to maintain model utility during unlearning while ensuring successful forgetting.

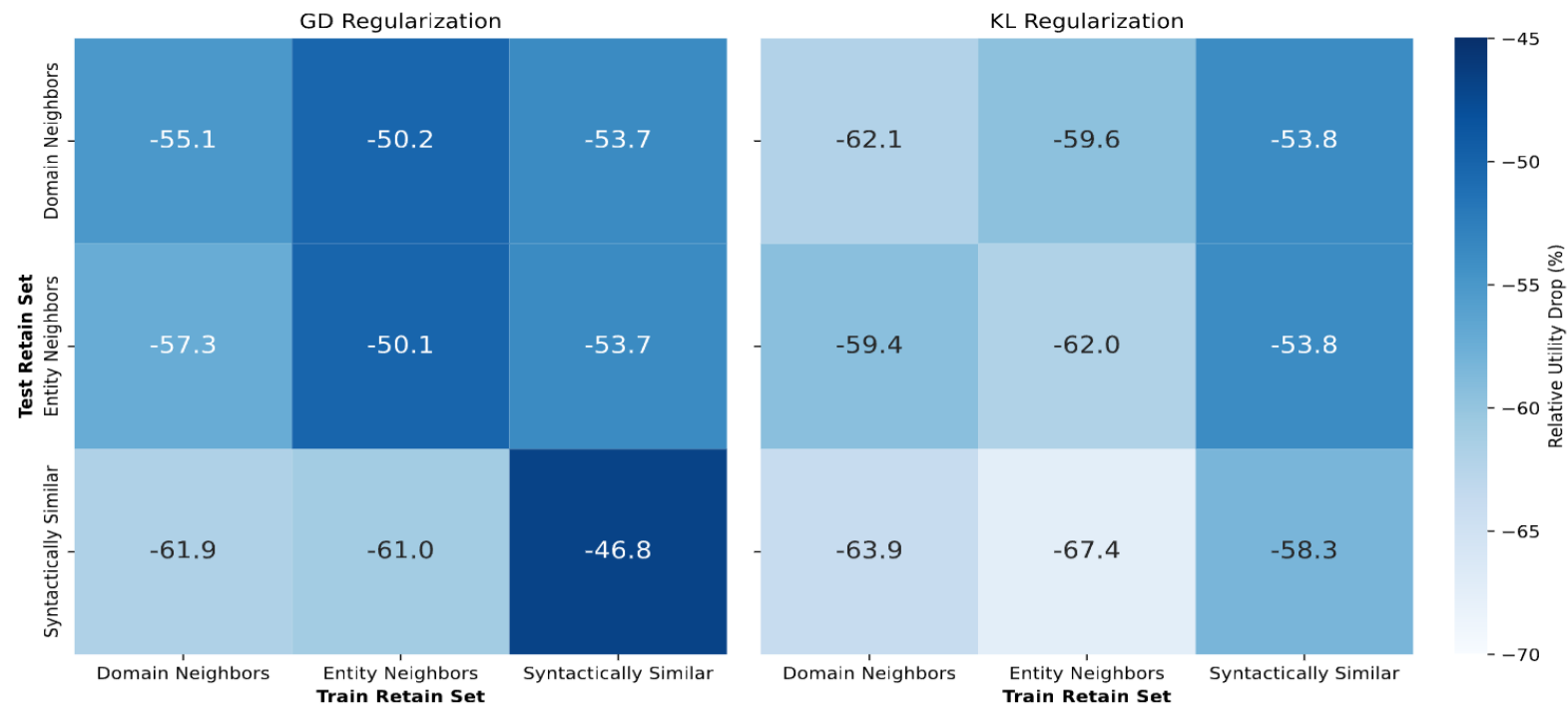
$$\min_{\theta} \mathcal{L}(\theta) = \min_{\theta} -\mathcal{L}_f(\theta) + \mathcal{L}_R(\theta; \mathcal{D}_R).$$

We test nine combinations of train and test retain sets (domain, entity, syntactically similar neighbor) using GD and KL regularization losses across various unlearning methods.



Optimal Neighbor Set for Effective Regularization

- Training with syntactically similar set consistently preserves performance on syntactically similar test sets, with significant performance gains.
- Training with syntactically similar set also maintains strong performance on Nentity and Ndomain, indicating robustness beyond syntactic similarity.



Conclusion

- Our study reveals that syntactic similarity, rather than domain or entity, predominantly drives utility degradation in retain sets.
- Incorporating syntactically similar neighbors into regularization strategies effectively mitigates performance loss, refining unlearning strategies for large language models.

Contact: hwanchang@cau.ac.kr



References

[1] Does Your LLM Truly Unlearn? Exploring the Recovery of Unlearned Knowledge.

<https://textify.ai/does-your-llm-truly-unlearn-exploring-the-recovery-of-unlearned-knowledge/>

[2] Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C. Lipton, and J. Zico Kolter.
TOFU: A Task of Fictitious Unlearning for LLMs.

